MILITARY SPECIFICATION

ANTENNA, TRANSMISSION LINE, AS-3437/BRC

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance, design, manufacture, and acceptance of the Antenna, Transmission Line, AS-3437/BRC, hereinafter referred to as the transmission line antenna.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-S-760	Strapping, Nonmetallic (And Connectors)

MILITARY

MIL-C-12000	Cable, Cord, And Wire, Electric; Packaging Of
MIL-S-13282	Silver And Silver Alloy
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For

NAVAL UNDERWATER SYSTEMS CENTER (NUSC), NEW LONDON LABORATORY (NLL)

NUSC-C-342/4141-279	Cable, Radio Frequency, Buoyant, General Specification

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-810C	Environmental Test Methods

2.1.2 Other Government drawings (DWGs). The following other Government DWGs form a part of this specification to the extent specified herein. Unless otherwsise specified, the issues shall be those in effect on the date of the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR-8111), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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DWGs

NUSC

NUSC/NLL 02378001	Cable Assembly, RF CG-3819/BRC And CG-3820/BRC (For Use With CG-3819/BRC Only)
NUSC/NLL 02379005	Pin Contact
NUSC/NLL 02387001	Adapter, Radio Frequency MX-10158/BRC
RE D 49485	Connector, UG-1820/U

(Copies of specifications, standards, and DWGs required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

DWGs

SPEARS ASSOCIATES, INCORPORATED (SAI)

SAI D185015	Cover, Test/Shipping
SAI D185049	Housing Assembly
SAI D211052	Connector, Plug, Electrical Assembly (Buoyant Cable)
SAI E185033	Connector, Plug, Electrical (Socket Contact) Cable Assembly R.F.
SAI E211026	R.F. Cable Assembly, CG-3820/BRC
SAI E211051	Antenna Transmission Line, AS-3437/BRC

(Copies of DWGs required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

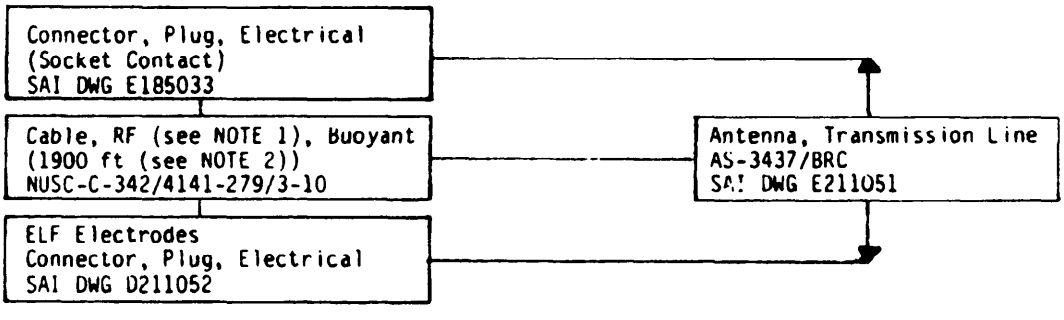
3. REQUIREMENTS

3.1 General. The transmission line antenna shall be in accordance with MIL-E-16400 to the extent specified herein.

3.1.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Description. The transmission line antenna shall consist of coaxial buoyant cable with a twisted pair transmission line and extremely low frequency (ELF) electrodes and connectors.

3.2.1 Transmission line antenna. The cable assembly shall consist of an assembly of components as shown in FIGURE 1.



NOTES: 1. Radio frequency
2. Feet

FIGURE 1. Components of transmission line antenna.

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3.3 Characteristics. Characteristics shall be as specified in 3.3.1 through 3.3.4.1.

3.3.1 Performance. The transmission line antenna shall contain a coaxial transmission line for conveying very low frequency (VLF) through high frequency (HF) signals from an attached antenna element through a coaxial transmission line contained within the cable and into a submarine. The cable shall also have a pair of ELF electrodes fabricated integrally with the cable jacket and exposed to sea water. A transmission line consisting of a twisted wire pair shall be incorporated within the cable and shall be used for conveying ELF signals received on the two outboard electrodes to ELF receiving equipment within the submarine.

3.3.2 Interface requirements. The electrical plug connector (socket contact) of the transmission line antenna shall mechanically and electrically mate with the RF Amplifier, AM-7074/BRC and the RF Adapter, MX-10158/BRC of Buoyant Cable Antenna Assembly, AS-3434(V)/BRC. The inboard connector of the transmission line antenna shall mechanically and electrically mate with the connector Rotary Coupler, CU-2269/BRC of Antenna Group, OE-315(V)/BRC. Electrical connections between the inboard connector and the transmission line antenna cable shall be as specified in TABLE I.

TABLE I. Inboard cable connector connections.

Pin indicators of inboard connector	Cable lead connections
A	Spare (not used)
C	White lead-twisted pair (inboard electrode)
B	Blue lead-twisted pair (outboard electrode)
D	Spare (not used)
E (outer conductor of coaxial contact)	Outer conductor coaxial line
F (pin of coaxial contact)	Center conductor coaxial line

3.3.3 ELF performance. The transmission line antenna shall provide the capability for receiving ELF signals when the Antenna Group, OE-315(V)/BRC is installed on SSN and SSBN submarines. ELF signals shall be received with a pair of electrodes which are fabricated as an integral part of the cable. The ELF performance specified in 3.3.3.1 and 3.3.3.2 shall be provided simultaneously with the VLF and low frequency (LF) and medium frequency (MF) and HF performance specified in 3.3.4 which is normally provided through use of other elements of the Antenna, AS-3434(V)/BRC.

3.3.3.1 ELF frequency range. The ELF antenna portion of the transmission line antenna shall provide ELF reception over the 10 hertz (Hz) to 200 Hz frequency range. The ELF signals shall be received on the ELF electrodes and passed through the cable to the output of the inboard connector.

3.3.3.2 ELF direct current (DC) resistance. When tested in accordance with 4.5.1, the transmission line antenna shall conform to the continuity and isolation resistance limits specified in TABLE II.

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TABLE II. Transmission line antenna DC resistance requirements.

Between points (see FIGURE 2)	DC resistance ohms per 100 ft	
F and H (continuity)	Not greater than 0.62	Center conductor coaxial
E and G (continuity)	Not greater than 0.70	Braid coaxial
B and J (continuity)	Not greater than 1.86	Twisted pair inboard electrode
C and I (continuity)	Not greater than 1.86	Twisted pair outboard electrode
F and E (isolation)	At least 1000 megohms	Coaxial center to braid
B and C (isolation)	At least 10 megohms	Inboard electrode to outboard electrode
C and E (isolation)	At least 10 megohms	Inboard electrode to coaxial braid
D and E (isolation)	At least 10 megohms	Inboard electrode to coaxial braid

3.3.4 VLF and LF and MF and HF performance. The transmission line antenna shall transfer VLF through HF signals from an antenna element attached to the outboard connector by a coaxial transmission line to the connector at the inboard end.

3.3.4.1 Coaxial line frequency range. The coaxial transmission line within the cable assembly shall operate over the 10 kilohertz to 30 megahertz frequency range.

3.4 Physical characteristics. Physical characteristics shall be as specified in 3.4.1 through 3.4.11.

3.4.1 General characteristics. The transmission line antenna shall be in accordance with this specification and the guidance of SAI DWG E211051.

3.4.2 Length. The length of the transmission line antenna from the outboard connector plug (socket contact) to the inboard connector shall be 1900 ft ±25 ft.

3.4.3 ELF electrodes. A pair of ELF electrodes shall be fabricated integrally with the cable assembly.

3.4.3.1 ELF electrode (outermost and innermost). The outermost electrode shall be 150 ft ±20 ft inboard from the outboard connector plug (socket contact). The innermost electrode shall be located 1000 ft ±20 ft inboard from the outermost electrode.

3.4.3.2 Electrode length. The portion of each electrode which is exposed to sea water shall be a minimum of 12 inches (in.) long.

3.4.3.3 Electrode construction. The electrode shall consist of number 18 Grade A, 99.9 percent pure annealed silver wire in accordance with MIL-S-13282, helically wound 11 turns per in. into a groove which is molded into the outer jacket of the cable. Both ends of each electrode shall be connected to their respective wires of the ELF twisted pair transmission line.

3.4.4 Special protection features. Special protection features shall be as specified in 3.4.4.1 and 3.4.4.2.

3.4.4.1 Water block. The transmission line antenna shall be water-blocked to prevent sea water flow axially through the cable when subjected to the hydrostatic pressure test of 4.5.2. Water blocking shall be accomplished within 3 ft ±1 ft of the inboard connector.

3.4.4.2 Twisted pair excision. A section of the twisted pair transmission line shall be excised and the remaining conductor pairs insulated to prevent contact with sea water and with each other when subjected to the hydrostatic pressure test of 4.5.2. The excision shall be made at a point between the outboard electrode and the antenna element connector. After excision, the jacket shall be remolded to prevent entry of sea water.

3.4.5 Marking. The transmission line antenna shall conform to the identification and marking requirements of DWGs, parts lists, specifications, and other documents specified on SAI DWG E211051.

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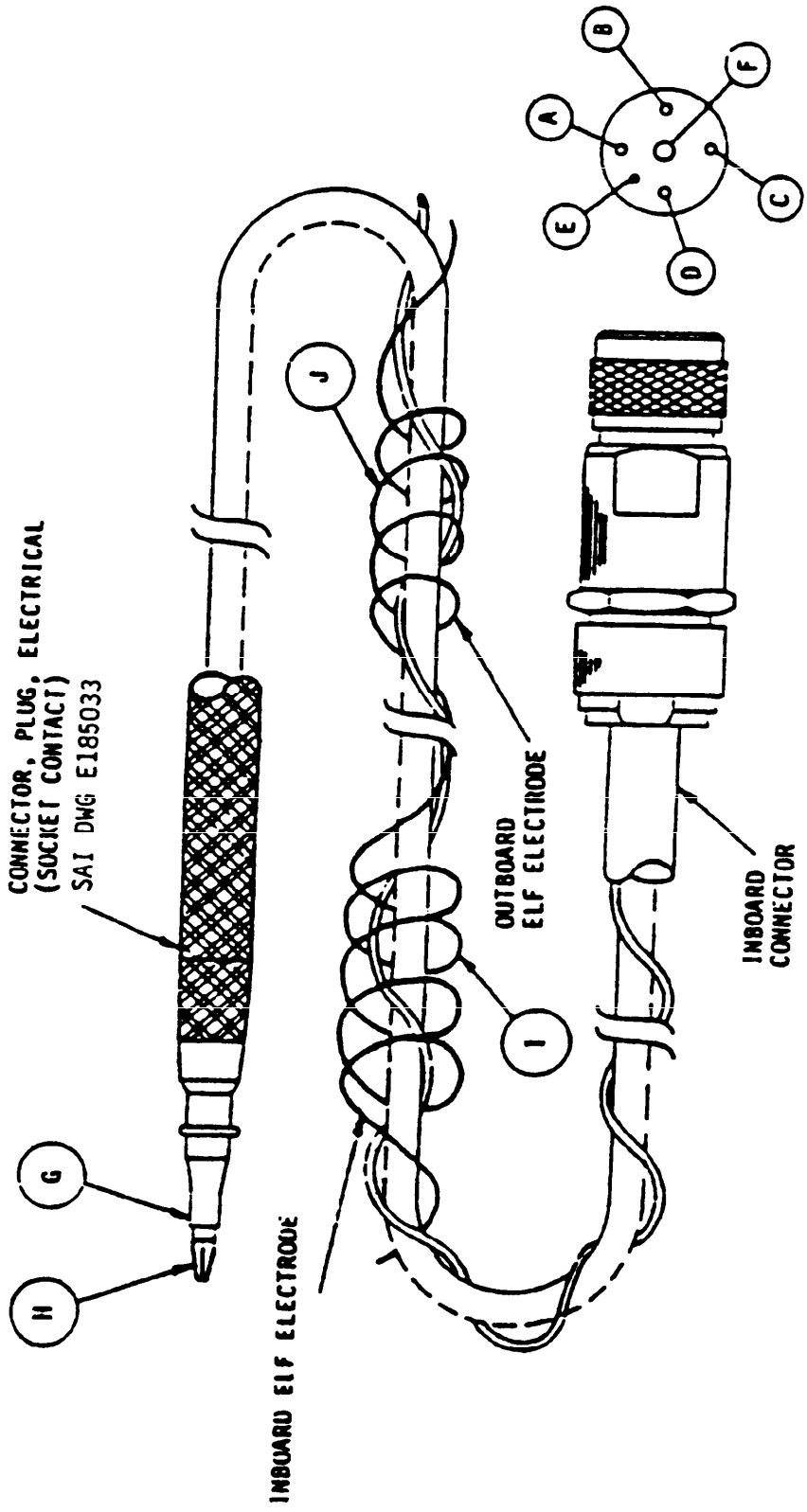


FIGURE 2. Test setup for ELF resistance test.

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3.4.5.1 Nameplate and product marking. The nameplate on the transmission line antenna shall be marked in accordance with MIL-STD-130.

3.4.6 Diameter and surface requirements. The transmission line antenna shall be fabricated using buoyant cable which has satisfactorily passed the requirements specified in NUSC-C-342/4141-279 for buoyant radio frequency (RF) cable. The outer diameter of any point along the complete antenna assembly, excluding the inboard connector, shall be 0.650 in., +0.020 in., -0.020 in. to ensure compatibility with the antenna transfer mechanism. Abrupt changes are identified as any variation of more than 0.010 in. diameter per in. in any direction along the surface of the cable. Abrupt change in diameter shall not be permitted. Any design cut into the cable jacket shall be stress-relieved to prevent cracking when subjected to repeated flexing. Design cuts or reduction in the cross sectional area shall not exceed 1.50 percent of the cross sectional area. The surface of the jacket shall be of uniform hardness and shall be free of major imperfections such as blow holes, cuts, valleys, and bruises.

3.4.7 Flexibility. When tested in accordance with 4.5.8, the transmission line antenna shall exhibit no physical damage or electrical degradation.

3.4.8 Pressure resistance. When tested under hydrostatic pressure in accordance with 4.5.2, the transmission line antenna shall show no evidence of damage or flooding.

3.4.9 Corrosion resistance. The coaxial center conductor, the twisted pair wires, all solder and weld joints, and internal connector hardware shall be positively protected from sea water contact.

3.4.10 Tensile strength. When tested in accordance with 4.5.11, the molded connector plug (socket contact) with suitable test fixtures attached, shall withstand static tensile loads along the plug's axis with no evidence of cracking or other damage. The maximum elongation of the molded section, including the separation of connector (metal section) from the buoyant cable, shall not exceed 2 percent when subjected to tensile loads as specified in TABLE V.

3.4.11 Durability. The transmission line antenna shall be compatible with the Antenna Transfer Assembly, AN/BRA-24( ). It shall be possible to deploy and fully retrieve the cable assembly at a pressure of 450 pounds per square inch (psi) utilizing the transfer assembly without damage to, or alteration of, the transfer assembly, and without damage to the antenna. No alteration or maintenance of the antenna shall be required for either deployment or retrieval.

3.5 Temperature. Temperature shall be as specified in 3.5.1 and 3.5.2.

3.5.1 Nonoperating temperature. The transmission line antenna shall withstand shipment and storage temperatures over the range of -40° Celsius (C) to +70°C.

3.5.2 Operating temperature. When tested in accordance with 4.5.3.2, the transmission line antenna shall operate at a temperature of -10°C.

3.6 Health and safety. Health and safety considerations require avoidance of toxicological substance such as encapsulants, adhesives, or preservatives which could contaminate the closed environment of a submarine. The materials specified in SAI DWG E211051 shall be acceptable. Any substitutes shall be approved by the Government.

3.7 Workmanship. The transmission line antenna shall be manufactured with care using the exact materials, dimensions, and surface finishes specified in SAI DWG E211051. Assembly shall ensure correct, arrangement, hookup, and fastening of all components. The molding procedure for the connector, plug, electrical (socket contact), and ELF electrodes is particularly critical and shall be performed under controlled time and temperature conditions. The completed assembly shall be clean and free of sand, grit, burrs, cuts, and gouges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall conform to all requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limted to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
c. Inspection of preparation for delivery (see 4.7)

4.3 First article inspection. Unless otherwise specified (see 6.2.1), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE III.

TABLE III. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection	
				Group A	Group B
DC resistance	3.3.3.2	4.5.1	X	X	
Pressure resistance	3.4.8	4.5.2	X	X	
Temperature:					
Nonoperating temperature	3.5.1	4.5.3.1	X		X
Operating temperature	3.5.2	4.5.3.2	X		X
Interface	3.3.2	4.5.4	X	X	
Physical characteristics	3.4.1, 3.4.2, 3.4.3.1, 3.4.3.2, 3.4.3.3	4.5.5	X	X	
Marking	3.4.5	4.5.6	X	X	
Identification and marking	3.4.5.1	4.5.6		X	
Diameter and surface	3.4.6	4.5.7	X	X	
Flexibility	3.4.7	4.5.8	X	X	
Corrosion resistance	3.4.9	4.5.9	X		
Durability	3.4.11	4.5.10	X		
Tensile strength	3.4.10	4.5.11	X		X
Workmanship	3.7	4.6	X	X	

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4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.2.2.

4.4.1 Production inspection (Group A). Production inspection shall be conducted on each transmission line antenna offered for delivery. Production inspection shall comprise such examinations and tests which will prove the workmanship and reveal omissions and errors of the production process such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, and tests which detect hidden defects of materials. Production inspection shall include the examinations and test specified in TABLE III, Group A.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.2.1. Production control inspection shall consist of the examinations and tests which encompass functional and performance tests throughout the entire range of operation, and tests to determine deviations in the processes and equipment required to fabricate the assembly. The tests shall be performed on the complete equipment as offered for delivery. Production control inspection shall include the examinations and tests specified in TABLE III, Group B.

4.4.2.1 Sampling plan for production control inspection. Equipment submitted for production control sampling test shall be selected by the Government quality assurance representative (QAR) and shall be representative of current production. One transmission line antenna shall be selected from each successive 50 production transmission line antennas produced.

4.4.2.2 Nonconforming production control sample units. If a sample unit fails the inspection specified in 4.4.2.1, the contractor shall immediately investigate the cause of failure and shall report to the QAR the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. No sample shall be delivered as a production item under the contract.

4.5 Test methods. To the extent specified herein, the examinations and tests specified in TABLE III shall be conducted in accordance with the test methods specified in MIL-E-16400.

4.5.1 DC resistance test. The transmission line antenna, as shown in FIGURE 2, shall be tested for continuity using a General Radio Type 1650-A impedance bridge, or equal. Insulation resistance readings shall be performed using a megohmeter (AN/PSM-25, or equal) which applies not more than 500 volts (V) to the circuit under test. The continuity and insulation resistance limits shall be in accordance with TABLE II.

4.5.2 Pressure resistance test. The transmission line antenna shall be tested for watertight integrity, using the test setup shown in FIGURE 3, as specified in a through e:

a. Place the cover, made in accordance with SAI DWG D185015, over the connector plug (socket contact) portion of the transmission line antenna as shown in SAI DWG E185033.

b. Subject the complete transmission line antenna and cover to hydrostatic pressure as specified in TABLE IV with the inboard connector end of the RF cable assembly in air at ambient pressure. The time to reach the required pressure shall be the minimum permitted by the equipment.

c. At the end of each pressure test interval, measure the insulation resistance using a megohmeter (AN/PSM-25, or equal) that applies no more than 500 V between test points. Measure between the center conductor and outer conductor of the coaxial portion of the inboard connector (test points C and D in FIGURE 2). The insulation resistance readings shall be in accordance with TABLE II.

d. Remove the transmission line antenna and cover from the pressure vessel and dry off the water on surface of the cable, connector, and cover.

e. Perform the surface examination in accordance with 4.5.7. Leave the cover in place on connector plug for protection during shipment (see 5.1).

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TABLE IV. Pressure resistance test.

First article inspection		Production inspection	
Pressure (psig) 1/	Duration (minutes)	Pressure (psig)	Duration (minutes)
0	5	0	5
450	25	450	25
0	5	0	5
450	25	450	25
0	5	0	5
450	25	450	25
0	5	0	5
450	4 hours	450	25
0	5	0	5
450	25	450	25
0	5	0	5
450	25	450	25
0	5	0	5
		450	25
		0	5
		450	25
		0	5

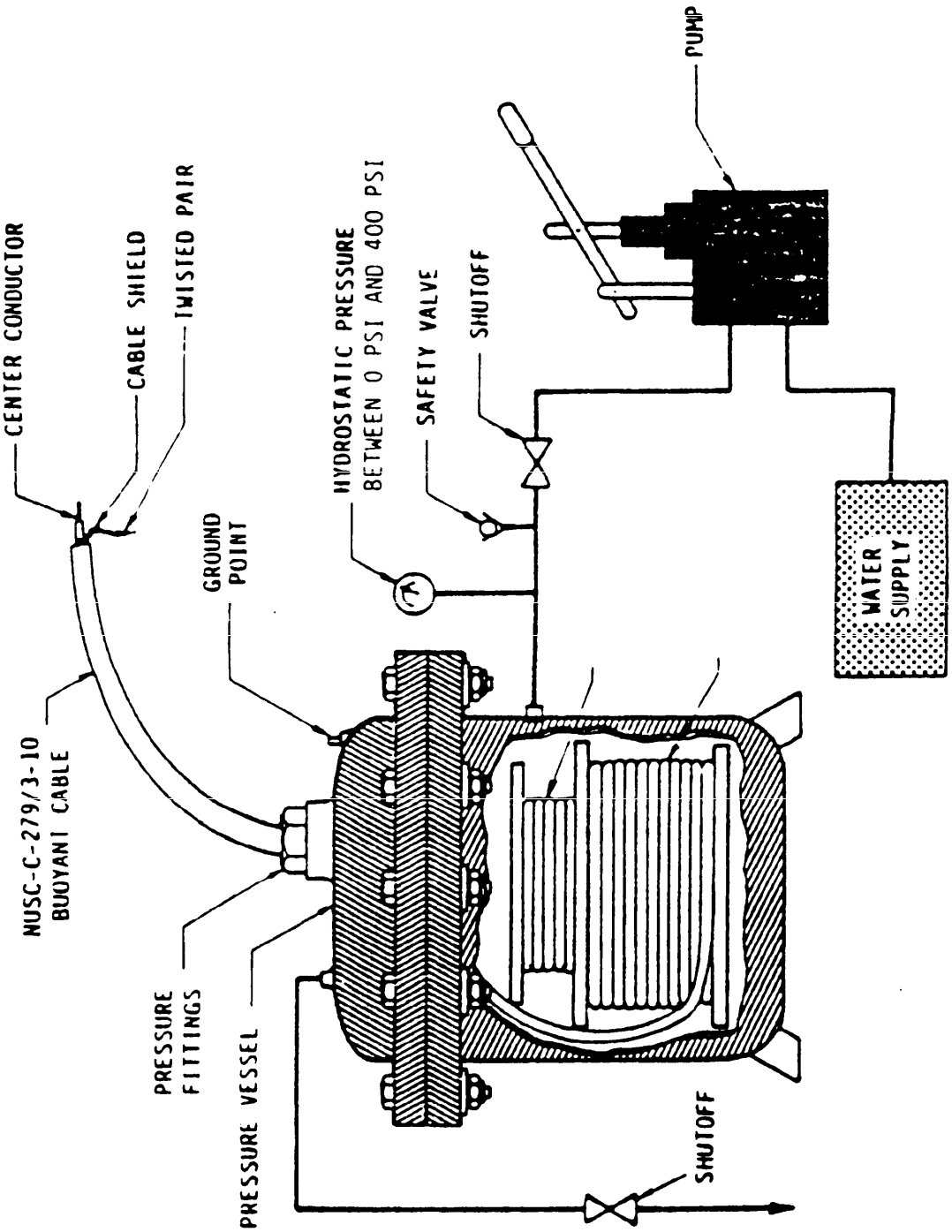
1/ Pound-force per square inch gage

4.5.3 Temperature test. Temperature test shall be conducted as specified in 4.5.3.1 and 4.5.3.2.

4.5.3.1 Nonoperating temperature. The transmission line antenna shall be formed into a coil having a maximum outside diameter of 36 in., minimum inside diameter of 15 in., and width less than 26 in. The transmission line antenna shall then be placed in an environmental chamber and conditioned for 8 hours at -40°C. The temperature shall be increased at a rate of 10°C per hour until the chamber reaches a temperature of +70°C. The temperature shall be held at 70°C for 8 hours and then the transmission line antenna shall be removed and inspected for damage. The transmission line antenna shall show no evidence of cracking or other damage. The performance shall be verified by conducting the DC resistance test of 4.5.1.

4.5.3.2 Operating temperature. The transmission line antenna, formed into a coil not over 36 in. in diameter, or on a reel as specified in 5.1, shall be conditioned in an environmental chamber for 12 hours at -10°C and, while still in the environmental chamber at -10°C, the connector plug (socket contact) portion shall be flexed through an S-bend path using the test setup shown in FIGURE 4 three times in each direction at 60 ft per minute (fpm). The electrical continuity shall be monitored continuously during the test. The connector plug shall be assembled with RF Adapter, MX-10158/BRC and the test cable shown in FIGURE 5 for this test. The performance shall be verified by inspecting the assembly for evidence of damage and by conducting the DC resistance test of 4.5.1.

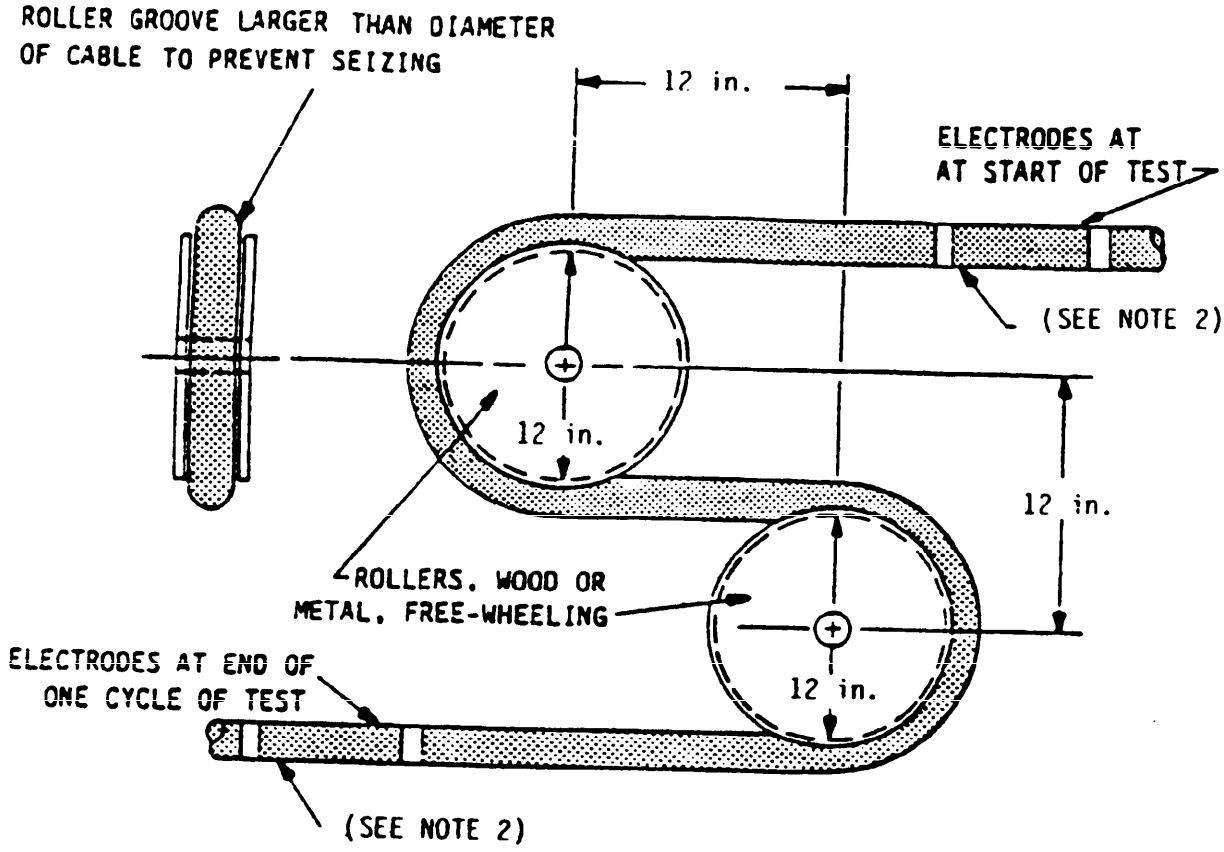
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NOTE: CONDUCTIVITY OF WATER SHALL BE AT LEAST 0.1 SIEMENS PER METER. ONE LEVEL TEASPOON OF BAKING SODA PER GALLON OF WATER SATISFIES THIS REQUIREMENT.

FIGURE 3. Test setup for transmission line antenna pressure test.

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NOTE: 1. USE SUFFICIENT TENSION TO BEND TRANSMISSION LINE ANTENNA COMPONENTS INTO CLOSE CONTACT WITH ROLLERS. ENSURE THAT TERMINATION TIP AND CONNECTOR PLUG, ELECTRICAL (PIN CONTACT) ARE HELD AGAINST ROLLERS, MAKING CONTACT THROUGH AT LEAST 180 DEGREES OF EACH ROLLER.

2. THE ADAPTER AND TEST CABLE ASSEMBLY SHOWN ON FIGURE 5 SHOULD BE ATTACHED TO THE ASSEMBLY TO FACILITATE TESTING.

FIGURE 4. Flexibility test setup.

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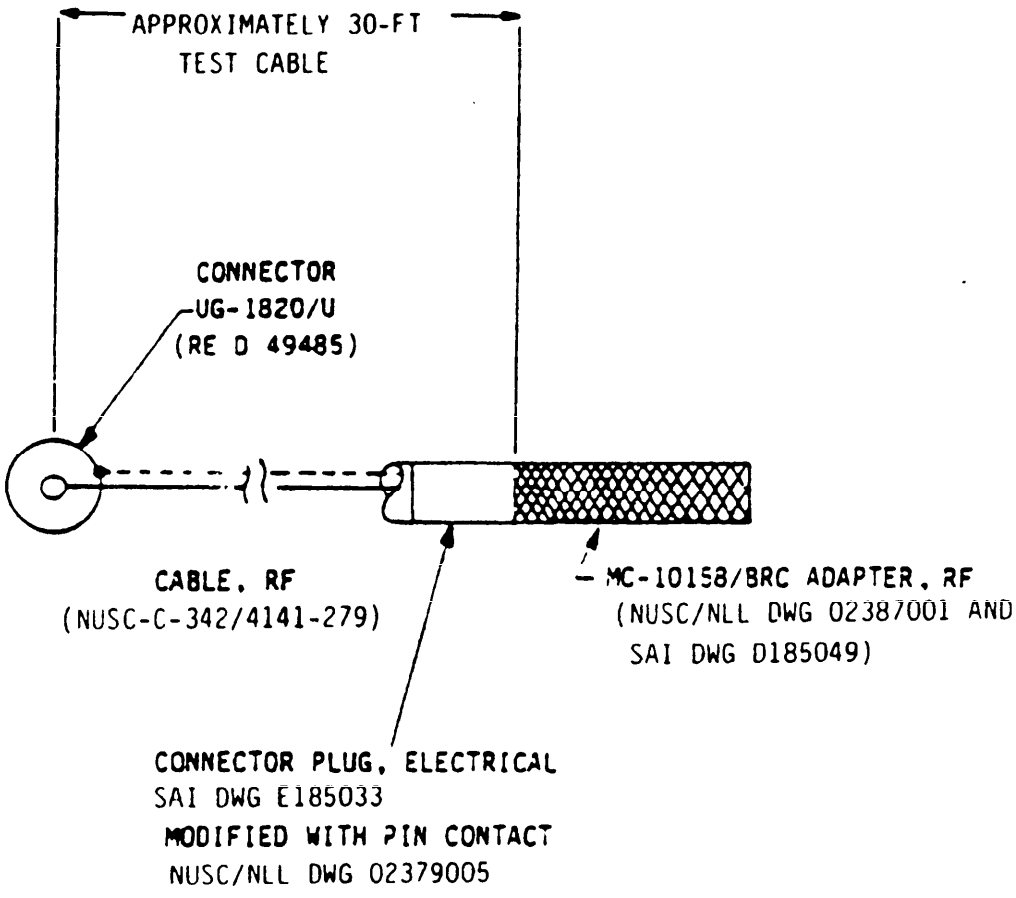


FIGURE 5. RF adapter, MX-10158/BRC, and test cable assembly.

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4.5.4 Interface examination. Appropriate visual and electrical examinations shall be performed to demonstrate compliance with the requirements of 3.3.2.

4.5.5 Dimensional verification. The length of each transmission line antenna and electrode spacing and construction shall be verified to demonstrate compliance with the requirements of 3.4.1, 3.4.2, 3.4.3.1, 3.4.3.2, and 3.4.3.3.

4.5.6 Marking verification. Tests shall consist of certifying mark locations and legibility and identification markings on the transmission line antenna as specified in 3.4.5 and 3.4.5.1.

4.5.7 Diameter and surface examination. Following visual examination in accordance with MIL-E-16400, the transmission line antenna, excluding the inboard connector, shall be passed through a sizing block having a hole of 0.671 in., +0.000 in., -0.001 in. diameter.

4.5.8 Flexibility test. The transmission line antenna, when assembled with mating component RF Adapter, MX-10158/BRC, and test cable including connector plug (pin contact), shall be flexed through an S-bend path in a test device made in accordance with FIGURE 4, three times in each direction at 60 fpm. The RF adapter and test cable assembly should be attached to the antenna element as shown in FIGURE 5 to facilitate testing. The electrical continuity shall be monitored continuously during the test.

4.5.9 Corrosion resistance test. Corrosion resistance test shall be conducted as specified in 4.5.9.1 and 4.5.9.2.

4.5.9.1 Transmission line antenna. The complete transmission line antenna, excluding the inboard connector, shall be immersed in sea water at a maximum pressure of 10 psig for a period of 1500 hours. The assembly shall show no degradation as demonstrated by electrical performance tests. No maintenance shall be required during the test.

4.5.9.2 Inboard connector. The inboard connector of the transmission line antenna shall be assembled on a cable sample and in the mated condition shall be subjected to the salt fog test specified in Method 509.1 of MIL-STD-810C. No corrosion products shall be visible upon completion of the test.

4.5.10 AN/BRA-24( ) durability test. The transmission line antenna shall be drawn 25 times in each direction through the Antenna Transfer Assembly, AN/BRA-24( ), using the test setup shown in FIGURE 6. The AN/BRA-24( ) shall contain water at a hydrostatic pressure of 50 psig to 120 psig. After completion of passage through the AN/BRA-24( ) for 25 cycles, a surface examination for evidence of cracking or other damage shall be performed in accordance with 4.5.7, the DC resistance test shall be performed in accordance with 4.5.1, and the marking examination shall be performed in accordance with 4.5.6.

4.5.11 Tensile test. The tensile strength of the molded connector plug (socket contact) of the transmission line antenna shall be tested by subjecting the plug to static tensile loads along the plug's axis in accordance with TABLE V. The test setup shall be similar to the one in FIGURE 7. The distance between point A (interface between metal part of connector plug and molded section of cable) and measurement point B (6 in. from point A) shall be measured under load at the beginning (initial 10-pound (lb) tensile load) and at the 400-lb tensile load during the fourth cycle. The elongation shall comply with the requirements of 3.4.10. The test sample shall be examined for any evidence of cracking or other damage. The flexibility test of 4.5.8 shall be performed.

4.6 Workmanship. The transmission line antenna shall be examined for workmanship to determine conformance to the surface examination requirements of 3.7 and the Workmanship paragraph of MIL-E-16400.

4.7 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

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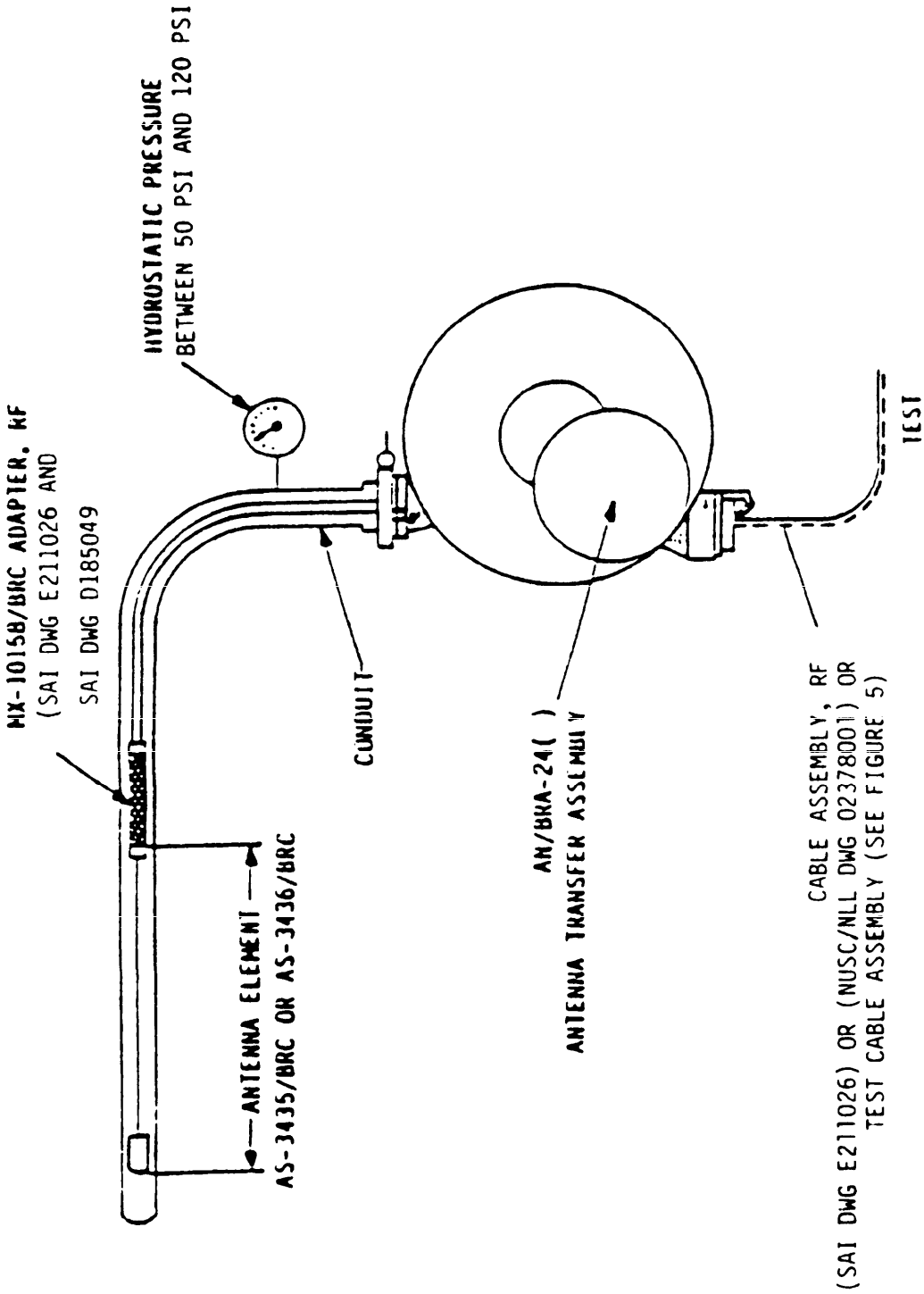


FIGURE 6. AN/BRA-24( ) durability test setup.

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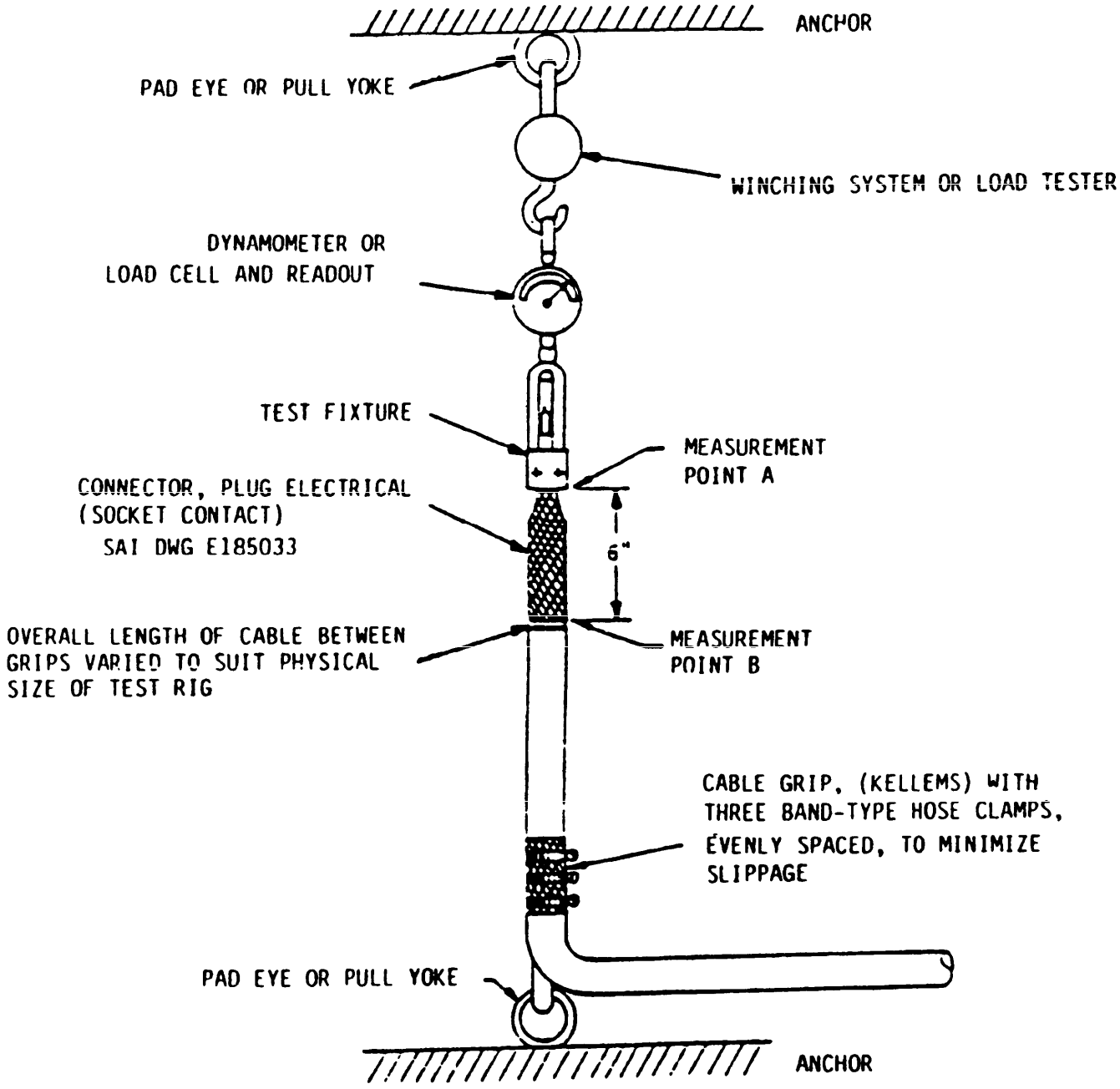


FIGURE 7. Tensile test setup.

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TABLE V. Tensile load test.

Connector plug (socket plug)	
Tensile load (lbs)	Duration (minutes)
10	Time required to perform reference measurement
400	55
10	5
400	55
10	5
400	55
10	5
400	55
10	5

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or purchase order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. No preservative shall be applied to the transmission line antenna. Packaging shall be on nonreturnable reels in accordance with level A of MIL-C-12000, except wood lagging shall be nominal 1 in. thick and lagging shall be strapped with three 0.625 in. by 0.030 in. nonmetallic straps conforming to PPP-S-760, Type II. Packaging shall afford protection against physical damage during shipment, handling, and uncovered storage. The outboard connector shall be accessible for checking continuity. The outboard connector (socket contact), cover assembly, and the inboard connector shall be sealed with a removable protective cap and shall be wrapped with cushioning material (Acrotube or Armoflex pipe insulation, or equal) for extra protection. The inboard connector shall be at the circumference of the transmission line section of the reel and shall be protected with a cap. Reels shall have the dimensions specified in a through c:

a. Hub not less than 18 in.
b. Width not more than 26 in.
c. Outside diameter not more than 36 in.

6. NOTES

6.1 Intended use. The transmission line antenna covered by this specification is intended for use by submarines as part of Buoyant Cable Antenna Assembly, AS-3434(V)/BRC.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. Levels of preservation, packaging, packing, and marking (see 5.1)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List, incorporated into the contract. When the provisions of DoD Federal Acquisition Regulations Supplement, Part 27, Sub-Part 27 410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs:

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Paragraph	Data requirement	Applicable DID
4.3	First Article Inspection Procedure	DI-T-4901
4.3	First Article Inspection Report	DI-T-4902

(DIDs related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L, Vol. II, AMSDL. Copies of DIDs required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Supersession data. This specification includes the requirements of ELEX-A-525, dated 18 August 1982.

Preparing activity:
NAVY - EC

(Project No. 5985-N525-2(EC))

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-28896	Antenna, Transmission Line, AS-3437/BRC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:

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a. Paragraph Number and Wording:

b. Recommended Wording:

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Custodians:

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Navy—SH

Air Force—11

International interest (see section 6)

Reviewer:

Army—EL, MI, MR, MU

Navy—SH

Air Force—11, 17, 69

User:

Army—GL

Navy—WP

Preparing activity:

Navy—SH

(Project 9330-0190)

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